Norm-4033 Eisenstein radix: selector proof, native costs, and limits

Question. Does the π=64+ω=65−τ Eisenstein radix improve scalar-local multiplication over the preceding norm-7009 path, and how does the result compare with OpenSSL's fixed-base secp256k1 multiplication?

Scope and instance. This study uses SEC 2 secp256k1 ($E: y^2 = x^3 + 7$, $p = 2^{256} - 2^{32} - 977$, subgroup order n=FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFEBAAEDCE6AF48A03BBFD25E8CD0364141). The full fixture has 12,288 scalar/point pairs. Its SHA-256 is:

a1c4964f174e36ea26f6a1897128f651a2ce7251e4bf4ffd532ff09221e721d3

It is an implementation and measurement study for this existing curve; it adds no curve, isogeny, or canonical graph edge.

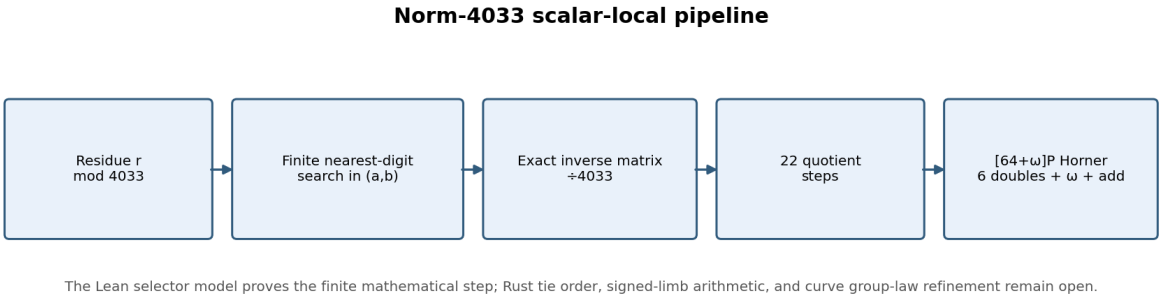


Figure 1: Norm-4033 recoding and evaluation pipeline

Selector and arithmetic. The hash-guarded overlay uses π=64+ω=65−τ of norm 4033. Its digit selector solves $a+65b \equiv r \pmod{4033}$, lifts the residue, and searches a five-by-five window around a rounded norm-minimizing center with deterministic tie breaks. The inverse matrix for multiplication by 65−τ produces quotient numerators $(62A-3B,\ A+65B)$, each divided exactly by 4033. The recoder uses 22 steps; the Horner action is $[64+\omega]P$, implemented with six doublings, one order-three x rotation, and one projective addition.

The source-bound selector auditor independently checks all 2,017 nonnegative signed residues against an exhaustive lattice minimum; negation covers the remaining residues for 4,033 total. It verifies the quotient congruence for every class, maximum selected-digit norm 1,323, and maximum coordinate magnitudes 63 and 42. This is a finite residue-domain result, not a source-level proof of the Rust fixed-width operations or group law.

Lean scope. The pinned radix module checks the norm, quotient adjugate, exact quotient under the selector congruence, and radix action. Radix4033Bounds.lean proves a uniform rounded-digit norm bound and a 22-step terminal table bound under explicit lattice-envelope and quotient hypotheses. Radix4033Selector.lean proves the rounded witness lies in the finite search and that the mathematical least-norm choice yields an exact quotient for every integer-pair state. I reran all three checkers with Lean 4.34.1; the dated receipts (radix, bounds, selector) record source hashes and `#print axioms`. The Rust tie order, signed-limb arithmetic, x rotation, and curve group-law refinement remain open.

Tier 1 comparison with norm-7009. On RunPod pod `thyvmdqmzsz5ik`, CPU 40, the one-scalar plan completed 96 arms and 48 pairs with all A/A and artifact checks passing. The primary manifest, rules, launch, and raw archive identify job 5198a67ca1fb418884feeb1585ad7d67. A separately frozen same-allocation replication with the same plan used job 85e158fb07c14ef78a803932b89da785; its launch, audit, and raw archive preserve the 96 arms and 48 pairs. The prepared 128-point panel is job 8fd7116208d94dee871413c6e33b4859, with its plan, rules, launch, and raw archive.

Workload	Norm-7009 mean	Norm-4033 mean	Qualified reference/candidate median	Paired bootstrap 95% interval
One scalar, initial panel	0.542 ms	0.471 ms	1.140×	[1.130, 1.144]
One scalar, same-allocation replication	0.542 ms	0.490 ms	1.132×	[1.117, 1.146]
128-point prepared stream	39.500 ms	38.959 ms	1.017×	[1.010, 1.019]

The independent auditor verified the source, binary, raw commands and outputs, pair records, independent scalar replay of measured points, artifact stability, and A/A controls. Each panel is Tier 1 evidence from one allocation; the one-scalar replication is not a second-allocation replication.

Comparison with OpenSSL. The OpenSSL baseline calls OpenSSL 3.0.13 `EC_POINT_mul` on the same scalars and points. Its timed interval includes multiplication and affine `x:y` serialization; fixture parsing, curve setup, one warm multiplication, and process launch are excluded. The custom path times scalar recoding, graph digit synthesis, Horner evaluation, and affine output. Thus the reference/candidate ratios below compare the declared application intervals, with the excluded OpenSSL warmups retained in the audit.

Workload	OpenSSL mean	Norm-4033 mean	Qualified OpenSSL/custom median	Paired bootstrap 95% interval
One scalar	0.220 ms	0.474 ms	0.460×	[0.458, 0.470]
128 points	27.653 ms	38.792 ms	0.714×	[0.712, 0.717]

These qualified ratios below 1 show the norm-4033 scalar-local path was slower than OpenSSL on these cold and batched workloads. Each panel completed 96 arms and 48 pairs with passing A/A controls. Jobs `efd-cff6ed47646598f58aed0e059bcf5` and `71ebf8c5348a487a9a505933bb892b93` are preserved by their one-scalar and 128-point launch records, with matching audits and raw archives. The auditor verifies both OpenSSL and candidate source/binary hashes and independently replays the measured outputs.

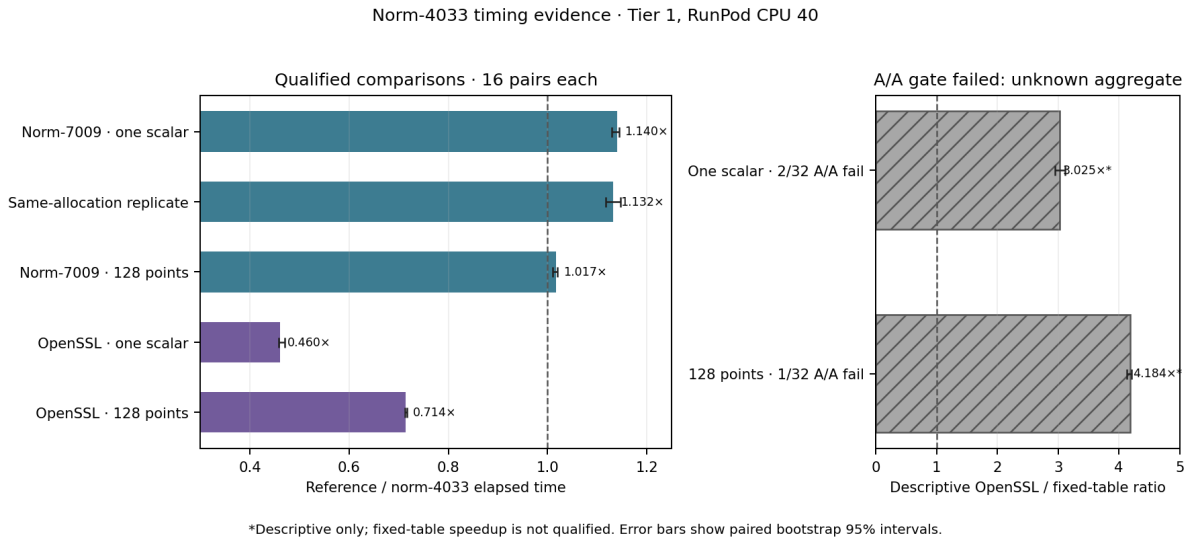


Figure 2: Qualified radix comparisons and A/A-gated fixed-table observations

Fixed-table engineering result. Reusing the norm-4033 digit and terminal tables reduces repeated point synthesis. The table retains 436,132 bytes; setup averaged 48.128 ms for the one-scalar panel and 49.497 ms for 128 points. The measured online means were 0.073 ms and 6.741 ms, with descriptive OpenSSL/reference ratios 3.025× and 4.184×. However, the frozen `[0.90,1.11]` calibration rule failed in 2 of 32 one-scalar A/A calibration pairs and 1 of 32 128-point A/A pairs. Therefore the aggregate speedup is **unknown**; the gray bars in the graph are descriptive only and not qualified comparisons. Their jobs were `1eab28e2edf04d8194bb174915625f37` and `2f8cb0ec64a147199e9e1b1d8d89da60`; the one-scalar and 128-point audits retain the failed controls and full intervals.

Fixture replays. The scalar-local candidate's A/A full-fixture replay (job `1d3cf18c0fc94bf3b4c9646d783156f2`) checked all 12,288 outputs in each arm at 3.652 s and 3.670 s. The fixed-table A/A replay (job `9beb2491d85c42da9652e15bb0cfe24e`) also checked all 12,288 outputs in each arm; its online intervals were 626.201 ms and 625.523 ms, with 47.165 ms and 47.191 ms preparation. Both have null speedup claims. Their auditors verify source, binary, fixture, outputs, raw records, and independent final-scalar replay.

Graph and proof status. No curve or route changed; I checked the canonical curve catalog and route records and found no graphable change. The norm-4033 problem record tracks the remaining selector-to-Rust and concrete point-map/group-law refinement. The measurements are workload- and allocation-specific, and the variable-digit path has not received a side-channel assessment. The editable figure source, SVG and 180-dpi PNG copies, and this report's PDF accompany the result.